PH-UY 2023: ELECTRICITY, MAGNETISM, & FLUIDS

MIDTERM EXAM Name _____

11/2/21 Univ ID _N_____ Section _____



BEFORE YOU DO ANYTHING ELSE….

1) **Print your full name neatly on your answer sheet in the Name area.**
2) **Write your course AND SECTION in the Subject area.**
3) **Write your N# in the first eight boxes under "STUDENT ID NUMBER".**
4) **Make sure you use #2 pencil. Pen will not register when graded!**

1. An object weighing 550 N is suspended from a rope and *completely submerged* in water (of density 1000 kg/m^3) as shown in the image. When the object is submerged in water, the tension in the rope is 20 N. What is the density of the object?



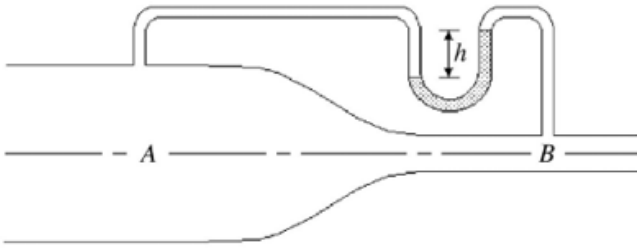
a) 1450 kg/m^3 b) 1120 kg/m^3 c) 1220 kg/m^3
d) 1322 kg/m^3 e) 1040 kg/m^3

2. Consider a circular stream of water leaving a faucet (see picture). Just after leaving the faucet, the stream flows with a velocity of 1.10 m/s and has a diameter of 2.00 cm. Find the diameter of the stream 1.00 m below the faucet.



a) 1.71 cm b) 0.982 cm c) 1.33 cm d) 1.56 cm e) 1.17 cm

3. Water flows in a horizontal pipe as shown in the figure. At point *A* the area is 25.0 cm^2 and the speed of the water is 2.00 m/s. At *B* the area is 16.0 cm^2. The fluid in the manometer (above the pipe) is mercury, which has a density of 13,600 kg/m^3. What is the manometer reading *h*?



a) 1.81cm b)1.39cm c) 2.16 cm d)5.01 cm e)3.28cm

4. A point charge of 8 μC is located on the x-axis at the point x = 3 m, and another point charge of -8 μC is located on the x-axis at the point x = -3 m. What is the magnitude of the force on a point charge of 5 μC on the y-axis at the point y = 5 m?

a) 2.98×10^{-2} N b) 2.39×10^{-2} N c) 1.96×10^{-2} N
d) 1.52×10^{-2} N e) 1.09×10^{-2} N

5. A 40 gram piece of styrofoam carries a net charge of 0.7 μC and floats in the air above a large horizontal charged dielectric sheet. Find the charge per unit area on the sheet.

a) 9.91 μC/m^2 b) 7.43 μC/m^2 c) 4.96 μC/m^2
d) 2.48 μC/m^2 e) 1.03 μC/m^2

6. A beam of protons traveling initially in the horizontal direction enters a region of space where there is an electric field of 350 V/m directed vertically downwards. After traveling a horizontal distance of 8 cm in this region, the protons are deflected a vertical distance of 3 mm. What is the initial speed of the protons when they enter the field region?

a) 3.78×10^5 m/s b) 5.86×10^5 m/s c) 1.89×10^5 m/s
d) 2.84×10^5 m/s e) 4.73×10^5 m/s

7. Electric charges are distributed uniformly with linear charge densities of +28 nC/m, -13 nC/m and +5.0 nC/m on three nonconducting threads located along the x-, y-, and z-axes respectively. What is the flux of the electric field through the surface of a sphere with a 6.0-cm radius centered at the origin?

a) 407 $N\cdot m^2/C$ b) 542 $N\cdot m^2/C$ c) 133 $N\cdot m^2/C$
d) 271 $N\cdot m^2/C$ e) 339 $N\cdot m^2/C$

8. A -2.3 nC charge is placed at the center of a thin conducting spherical shell of a 3.1 cm radius carrying a +1.1 nC charge, which is concentric with another thin conducting spherical shell of a 8.1-cm radius carrying a -6.2 nC charge. Determine the magnitude of the electric field at a distance of 4.0 cm from the surface of the inner sphere.

a) 2140 N/C b) 1605 N/C c) 892 N/C d) 357 N/C e) 981 N/C

9. Two 7.0 μC point charges are held fixed at the points (0 cm, 4 cm) and (0 cm, -4 cm). A third particle with a charge of 1.3 μC and mass 2.0 g is released from rest from the point (3 cm, 0 cm), what will its speed be when it is far from the origin?

a) 57.2 m/s b) 126 m/s c) 84.0 m/s d) 48.2 m/s e) 103 m/s

10. Four identical point charges +4.0 μC each are located at the corners of a square which has 25-cm sides. How much work was required to assemble this charge arrangement if the charges were initially very far from each other?

a) 9.54 J b) 8.22 J c) 7.01 J d) 4.87 J e) 3.12 J